

**Q1 - 25 July - Shift 1**

The total number of functions,

$$f: \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4, 5, 6\}$$

such that  $f(1) + f(2) = f(3)$ , is equal to :

(A) 60

(B) 90

(C) 108

(D) 126

Space for your notes:

**Q2 - 25 July - Shift 2**

The number of bijective functions  $f: \{1, 3, 5, 7, \dots, 99\} \rightarrow \{2, 4, 6, 8, \dots, 100\}$ , such that  $f(3) \geq f(9) \geq f(15) \geq f(21) \geq \dots \geq f(99)$ , is

(A)  ${}^{50}P_{17}$

(B)  ${}^{50}P_{33}$

(C)  $33! \times 17!$

(D)  $\frac{50!}{2}$

Space for your notes:

**Q3 - 25 July - Shift 2**

Let  $f(x)$  be a quadratic polynomial with leading coefficient 1 such that  $f(0) = p$ ,  $p \neq 0$  and

$$f(1) = \frac{1}{3}. \text{ If the equation } f(x) = 0 \text{ and } f(f(f(f(x)))) = 0$$

have a common real root, then  $f(-3)$  is equal to.....

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**Q4 - 26 July - Shift 1**

Questions

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Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be a continuous function such that  $f(3x) - f(x) = x$ . If  $f(8) = 7$ , then  $f(14)$  is equal to :

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- (A) 4 (B) 10  
(C) 11 (D) 16

Q5 - 27 July - Shift 1

Let  $f, g: \mathbb{N} - \{1\} \rightarrow \mathbb{N}$  be functions defined by  $f(a) = \alpha$ , where  $\alpha$  is the maximum of the powers of those primes  $p$  such that  $p^\alpha$  divides  $a$ , and  $g(a) = a + 1$ , for all  $a \in \mathbb{N} - \{1\}$ . Then, the function  $f + g$  is

Space for your notes:

- (A) one-one but not onto  
(B) onto but not one-one  
(C) both one-one and onto  
(D) neither one-one nor onto

Q6 - 27 July - Shift 2

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The domain of the function

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$$f(x) = \sin^{-1}[2x^2 - 3] + \log_2 \left( \log_{\frac{1}{2}} \left( x^2 - 5x + 5 \right) \right),$$

where  $[t]$  is the greatest integer function, is :

(A)  $\left( -\sqrt{\frac{5}{2}}, \frac{5-\sqrt{5}}{2} \right)$  (B)  $\left( \frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2} \right)$

(C)  $\left( 1, \frac{5-\sqrt{5}}{2} \right)$  (D)  $\left[ 1, \frac{5+\sqrt{5}}{2} \right)$

**Q7 - 27 July - Shift 2**

The number of functions  $f$ , from the set

Space for your notes:

$$A = \{x \in \mathbb{N} : x^2 - 10x + 9 \leq 0\}$$
 to the set

$$B = \{n^2 : n \in \mathbb{N}\}$$
 such that  $f(x) \leq (x-3)^2 + 1$ , for

every  $x \in A$ , is \_\_\_\_\_.

**Q8 - 28 July - Shift 1**

Considering only the principal values of the inverse trigonometric functions, the domain of the

function  $f(x) = \cos^{-1}\left(\frac{x^2 - 4x + 2}{x^2 + 3}\right)$  is :

(A)  $\left(-\infty, \frac{1}{4}\right]$  (B)  $\left[-\frac{1}{4}, \infty\right)$

(C)  $\left(-\frac{1}{3}, \infty\right)$  (D)  $\left(-\infty, \frac{1}{3}\right]$

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**Q9 - 28 July - Shift 1**

Let  $\alpha, \beta$  and  $\gamma$  be three positive real numbers. Let

$f(x) = \alpha x^5 + \beta x^3 + \gamma x, x \in \mathbb{R}$  and  $g: \mathbb{R} \rightarrow \mathbb{R}$  be

such that  $g(f(x)) = x$  for all  $x \in \mathbb{R}$ . If  $a_1, a_2, a_3, \dots, a_n$  be in arithmetic progression with mean zero, then

the value of  $f\left(g\left(\frac{1}{n} \sum_{i=1}^n f(a_i)\right)\right)$  is equal to :

(A) 0 (B) 3  
(C) 9 (D) 27

Space for your notes:

**Q10 - 29 July - Shift 2**

The number of elements in the set

$S = \left\{x \in \mathbb{R} : 2\cos\left(\frac{x^2 + x}{6}\right) = 4^x + 4^{-x}\right\}$  is:

(A) 1 (B) 3  
(C) 0 (D) infinite

Space for your notes:

**Q11 - 29 July - Shift 2**

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The domain of the function

Space for your notes:

$$f(x) = \sin^{-1} \left( \frac{x^2 - 3x + 2}{x^2 + 2x + 7} \right) \text{ is :}$$

(A)  $[1, \infty)$

(B)  $(-1, 2]$

(C)  $[-1, \infty)$

(D)  $(-\infty, 2]$

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## Answer Key

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**Q1 (B)**

**Q2 (B)**

**Q3 (25)**

**Q4 (B)**

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**Q5 (D)**

**Q6 (C)**

**Q7 (1440)**

**Q8 (B)**

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**Q9 (A)**

**Q10 (A)**

**Q11 (C)**

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**Q1 (B)**

$$A = \{1, 2, 3, 4\}$$

$$B = \{1, 2, 3, 4, 5, 6\}$$

Here  $f(3)$  can be 2, 3, 4, 5, 6

$$f(3) = 2, (f(1), f(2)) \rightarrow (1,1) \rightarrow 6 \text{ cases}$$

$$f(3) = 3, (f(1), f(2)) \rightarrow (1,2), (2,1)$$

$$\rightarrow 2 \times 6 = 12 \text{ cases}$$

$$f(3) = 4, (f(1), f(2)) \rightarrow (1,3), (3,1), (2,2)$$

$$\rightarrow 3 \times 6 = 18 \text{ cases}$$

$$f(3) = 5, (f(1), f(2)) \rightarrow (1,4), (4,1), (2,3), (3,2)$$

$$\rightarrow 4 \times 6 = 24 \text{ cases}$$

$$f(3) = 6, (f(1), f(2)) \rightarrow (1,5), (5,1), (2,4), (4,2), (3,3)$$

$$\rightarrow 5 \times 6 = 30 \text{ cases}$$

$$\text{Total number of cases} = 6 + 12 + 18 + 24 + 30 = 90$$

**Q2 (B)****(B)****Q3 (25)****(25)****Q4 (B)**



$$f(x) - f(x/3) = x/3$$

$$f(x/3) - f(x/3^2) = x/3^2$$

.... on adding

$$f(x) - \lim_{n \rightarrow \infty} f\left(\frac{x}{3^n}\right) = x\left(\frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^n} + \dots\right)$$

$$f(x) - f(0) = \frac{x}{2}$$

$$f(8) = 7 ; f(0) = 3$$

$$f(x) = x/2 + 3$$

$$f(14) = 10$$

**Q5 (D)**



$$f: \mathbb{N} - \{1\} \rightarrow \mathbb{N} \quad f(a) = \alpha$$

Where  $\alpha$  is max of powers of prime  $P$  such that  $p^\alpha$  divides  $a$ . Also  $g(a) = a + 1$

$$\therefore f(2) = 1 \quad g(2) = 3$$

$$f(3) = 1 \quad g(3) = 4$$

$$f(4) = 2 \quad g(4) = 5$$

$$f(5) = 1 \quad g(5) = 6$$

$$\Rightarrow f(2) + g(2) = 4$$

$$(f(3) + g(3)) = 5$$

$$f(4) + g(4) = 7$$

$$f(5) + g(5) = 7$$

$\therefore$  Many one  $f(x) + g(x)$  does not contain 1

$\Rightarrow$  into function

$\therefore$  Ans. (D) [neither one-one nor onto]

**Q6 (C)**

$$f(x) = \sin^{-1}[2x^2 - 3] + \log_2 \left( \log_{\frac{1}{2}}(x^2 - 5x + 5) \right)$$

$$P_1: -1 \leq [2x^2 - 3] < 1$$

$$\Rightarrow -1 \leq 2x^2 - 3 < 1$$

$$\Rightarrow 2 < 2x^2 < 5$$

$$\Rightarrow 1 < x^2 < \frac{5}{2}$$

$$\Rightarrow P_1: x \in \left(-\sqrt{\frac{5}{2}}, -1\right) \cup \left(1, \sqrt{\frac{5}{2}}\right)$$

$$P_2: x^2 - 5x + 5 > 0$$

$$\Rightarrow \left(x - \left(\frac{5 - \sqrt{5}}{2}\right)\right) \left(x - \left(\frac{5 + \sqrt{5}}{2}\right)\right) > 0$$

$$P_3: \log_{\frac{1}{2}}(x^2 - 5x + 5) > 0$$

$$\Rightarrow x^2 - 5x - 5 < 1$$

$$\Rightarrow x^2 - 5x + 4 < 0$$

$$\Rightarrow P_3: x \in (1, 4)$$

$$\text{So, } P_1 \cap P_2 \cap P_3 = \left(1, \frac{5 - \sqrt{5}}{2}\right)$$

**Q7 (1440)**

$$(x^2 - 10x + 9) \leq 0 \Rightarrow (x - 1)(x - 9) \leq 0$$

$$\Rightarrow x \in [1, 9] \Rightarrow A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$f(x) \leq (x - 3)^2 + 1$$

$$x = 1 : f(1) \leq 5 \Rightarrow 1^2, 2^2$$

$$x = 2 : f(2) \leq 2 \Rightarrow 1^2$$

$$x = 3 : f(3) \leq 1 \Rightarrow 1^2$$

$$x = 4 : f(4) \leq 2 \Rightarrow 1^2$$

$$x = 5 : f(5) \leq 5 \Rightarrow 1^2, 2^2$$

$$x = 6 : f(6) \leq 10 \Rightarrow 1^2, 2^2, 3^2$$

$$x = 7 : f(7) \leq 17 \Rightarrow 1^2, 2^2, 3^2, 4^2$$

$$x = 8 : f(8) \leq 26 \Rightarrow 1^2, 2^2, 3^2, 4^2, 5^2$$

$$x = 9 : f(9) \leq 37 \Rightarrow 1^2, 2^2, 3^2, 4^2, 5^2, 6^2$$

Total number of such function

$$= 2(6!) = 2(720) = 1440$$

Q8 (B)

$$\left| \frac{x^2 + 4x + 2}{x^2 + 3} \right| \leq 1$$

$$\Leftrightarrow (x^2 - 4x + 2)^2 \leq (x^2 + 3)^2$$

$$\Leftrightarrow (x^2 - 4x + 2)^2 - (x^2 + 3)^2 \leq 0$$

$$\Leftrightarrow (2x^2 - 4x + 5)(-4x - 1) \leq 0$$

$$\Leftrightarrow -4x - 1 \leq 0 \rightarrow x \geq -\frac{1}{4}$$

Q9 (A)

Consider a case when  $\alpha = \beta = 0$  then

$$f(x) = yx$$

$$g(x) = \frac{x}{y}$$

$$\frac{1}{n} \sum_{i=1}^n f(a_i) \Rightarrow \frac{y}{n} (a_1 + a_2 + \dots + a_n)$$

$$= 0$$

$$\Rightarrow f(g(0)) \Rightarrow f(0)$$

$$\Rightarrow 0$$

**Q10 (A)**

$$2 \cos \left( \frac{x^2 + x}{6} \right) = 4^x + 4^{-x}$$

L.H.S  $\leq 2$  & R.H.S.  $\geq 2$

Hence L.H.S = 2 & R.H.S = 2

$$2 \cos \left( \frac{x^2 + x}{6} \right) = 2 \quad 4^x + 4^{-x} = 2$$

Check  $x = 0$  Possible hence only one solution.

**Q11 (C)**

$$f(x) = \sin^{-1}\left(\frac{x^2 - 3x + 2}{x^2 + 2x + 7}\right) \text{ Domain}$$

$$\frac{x^2 - 3x + 2}{x^2 + 2x + 7} \geq -1 \text{ and } \frac{x^2 - 3x + 2}{x^2 + 2x + 7} \leq 1$$

$$2x^2 - x + 9 \geq 0 \text{ and } 5x \geq -5 \Rightarrow x \geq -1$$

$$x \in \mathbb{R}$$

$$\text{Hence Domain } x \in [-1, \infty)$$